MACHINE LEARNING PROJECT

Star Classification

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This report presents a machine learning analysis for classifying astronomical objects from Sloan Digital Sky Survey data into galaxies, stars, and quasars. It compares several classification approaches and evaluates their ability to generalize to unseen observations.

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Business Understanding

The project focuses on the analysis of a dataset for the classification of astronomical objects, specifically stars, galaxies, and quasars. The classification is based on the main spectral characteristics of the objects, using data collected by the SDSS (Sloan Digital Sky Survey).

The main objective of the analysis is to create a predictive model capable of classifying detected astrophysical objects by using specific spectral features. In addition, the project aims to achieve further benefits, including:

- Greater operational efficiency, through the reduction of analysis time and of the costs associated with managing large datasets, by computerizing the process.
- Better analysis of astronomical data, allowing for more accurate classification.
- Greater efficiency in scientific discovery, thanks to the possibility of identifying new astronomical objects with higher precision.

To achieve these objectives, the original dataset is divided into three subsets with different proportions:

- **Training set**: used to train the model.
- **Validation set**: useful for optimizing parameters and preventing overfitting.
- **Test set**: used to evaluate model performance.

The analysis compares different classification models, including:

- **Discriminant Analysis**:
  - **LDA (Linear Discriminant Analysis)**: assumes that the classes have normal distributions with the same covariance matrix, producing linear decision boundaries.
  - **QDA (Quadratic Discriminant Analysis)**: allows each class to have a different covariance matrix, producing quadratic decision boundaries.
- **K-Nearest Neighbors (KNN)**: a method based on the proximity of points in the feature space.
- **Decision Trees**:
  - They iteratively split the dataset according to the values of the predictive variables.
  - They are interpretable and do not require categorical variables to be transformed.
  - They can suffer from overfitting if they are not pruned correctly.
  - More advanced versions, such as Random Forest or Gradient Boosting Trees, can improve performance by reducing variance.

The objective is to identify the model that handles the data best, balancing the trade-off between bias and variance, avoiding misspecification and overfitting problems, and ensuring strong generalization ability.

The key questions guiding the analysis are:

1. Which variables influence the classification the most?
2. How well is the model able to interpret the phenomenon under analysis?
3. Are there common anomalous values across all variables that could influence the classification?
4. Are there underrepresented classes that could lead to the accuracy paradox?

Data Understanding

The dataset under analysis contains **100,000 astronomical observations** collected by the **Sloan Digital Sky Survey (SDSS)** and describes them through **17 numerical variables**. The last variable in the dataset is `class`, which is used to classify the observations into three categories: *Galaxy, Star, Quasar*.

ID variables

- `obj_ID`: Unique identifier of the object in the catalog.
- `spec_obj_ID`: Unique identifier for spectroscopic objects.
- `run_ID`: Identifies the specific scan.
- `rerun_ID`: Indicates how the image was processed.
- `cam_col`: Camera column used in the scan.
- `field_ID`: Field number used to identify the observed area.
- `plate`: ID of the plate that captured the spectrum.
- `MJD`: Modified Julian Date, indicating the time of the observation.
- `fiber_ID`: Identifies the optical fiber used to capture the object's light.

These ID variables are identifiers of the techniques and instruments used for processing and scanning the objects, such as plates, optical fibers, or the date on which the observation was carried out.

Spatial coordinate variables

- `delta`: Declination (DEC) in degrees.
- `alpha`: Right Ascension (RA) in degrees.

Celestial coordinates are used to identify the position of stars and other celestial objects on the celestial sphere. They use the celestial equator, hour circles or meridians, and the First Point of Aries (gamma), namely the point where the Sun passes from the southern celestial hemisphere to the northern one, as reference points. For additional context, declination is sometimes replaced by polar distance (p), which is the angular distance of the object from the north celestial pole and ranges from 0 degrees to 180 degrees. Since the angles are complementary, p + delta = 90 degrees.

Photometric filter variables

- `u`: Near-ultraviolet filter (355 nm).
- `g`: Green filter (477 nm).
- `r`: Red filter (622 nm).
- `i`: Near-infrared filter (763 nm).
- `z`: Infrared filter (893 nm).

A star emits a flux of electromagnetic radiation that depends on its surface temperature. This radiation is perceived by the human eye as visible light, which determines the color of the star as it appears to us. However, the light emitted by a star has a continuous spectrum that extends over a wide range of frequencies, and analyzing this spectrum makes it possible to understand the behavior of the celestial body.

Photometric filters are used to isolate light radiation across the different bands. These are carefully calibrated filters that make the percentage of luminous flux recorded at each frequency comply with a predefined standard. There are many photometric systems; the one considered in this dataset is the SDSS system, which uses the five optical bands described above.

Shift variable

- `redshift`: A phenomenon whereby the light emitted by objects moving away has a longer wavelength than it had at emission.

The English term “redshift” comes from the fact that the light moves toward the red end of the visible spectrum. The opposite case is called “blueshift”, where the wavelength decreases when the light moves toward an observer.

The observation technique used in this dataset is photometry, which is simpler but leads to a rather approximate estimate of redshift. The optimal alternative would be to evaluate redshift through spectroscopy, which is a more complex technique because it is negatively affected by factors such as lack of light and less professional instruments. In general, spectroscopy is the most precise method for measuring redshift, but when spectroscopy cannot be used, photometry still provides a valid estimate.

Target variable

- `class`: Indicates the category of the object (*star, galaxy, quasar*).

A brief description of these three categories is useful for understanding their meaning:

- A **galaxy** is a gravitationally bound system of stars and other forms of matter.
- A **star** is a celestial body made of plasma and high-temperature gas.
- A **quasar** is an extragalactic object that emits an enormous amount of energy.

Dataset presentation

In the dataset, each observation is unique at the `spec_obj_ID` level, but some of the observed objects are repeated at the `obj_ID` level and therefore are not unique objects. For this reason, repeated observations were removed from the dataset.

The dataset is divided into three subsets: a training set, which is used in the first phase to train the models and understand the relationships among variables (80% of the total data); a validation set, which is used to estimate model parameters and evaluate analysis quality (45% of the training set); and a test set (20% of the total data), which is used to evaluate the final performance of the selected model and its ability to generalize to new data.

The decision to include a validation set comes from the need to limit underfitting and overfitting effects that could arise during the analysis. Through this split, the project evaluates several classification models: K-Nearest Neighbors (KNN), multinomial regression, classification decision trees, Linear Discriminant Analysis, and Quadratic Discriminant Analysis, with the objective of identifying the method that best classifies the available data.

In addition to the object IDs mentioned above, other variables represent identifiers of the techniques and instruments used for processing and scanning the objects. From an analytical point of view, these variables are not particularly meaningful; moreover, they take values over very wide and highly different ranges, making them difficult to manage within the dataset. The only ID variable with a limited number of values is `cam_col`, which can be treated as a categorical variable with six levels and can therefore be handled relatively easily. For this reason, `cam_col` was kept in the final dataset, while the other ID variables were removed.

Data Pre-Processing

For the analysis, it is useful for the variables not to be correlated with each other, because this implies that each variable contributes distinct information to the final dataset and that there is no redundancy in the information carried by the variables. The observed result is therefore considered positive, and the analysis proceeds accordingly.

A previous analysis of the dataset through boxplots showed that the `redshift` variable contains less pronounced outliers than the other variables. The approach used is a logarithmic transformation. A value of +1 is added inside the logarithm to handle negative values; in the case of `redshift`, the minimum value was -0.09959, so adding 1 makes it mathematically possible to apply the logarithm. This transformation is included in the analysis because it makes the variable distribution more homogeneous and makes the outliers easier to manage, as they become more clearly marked in the tails of the distribution.

To bring all variables into a single variance range, the data are normalized. This limits the value ranges to [0,1] without changing the overall shape of the distributions. This choice is motivated by an initial analysis of the dataset using `summary()`, which showed that the individual columns take values over wide and very different ranges.

The graph shows the frequency distributions of the numerical variables under analysis, conditional on class. The most discriminating variable is the transformed `redshift`, which is highly informative in distinguishing the classes. The variables `alpha` and `delta` show a similar distribution, differing mainly in their frequency peaks. The wider and more marked distribution for the `GALAXY` class is considered understandable, given the larger number of observations in that class.

The variables identifying photometric magnitudes again show class-level differences, although these are less pronounced than in the distribution of the transformed `log_shift`. Specifically, the `GALAXY` label shows a bimodal pattern, the `QSO` label shows an asymmetric distribution, and the `STAR` label shows a flatter distribution across these five variables.

The `Galaxy` class is more represented than the others. For this reason, it was subject to undersampling in order to make the distribution of observations across the three classes more homogeneous. Even after this operation, the context remains one of imbalanced classes, because one class is still much larger than the others. This must therefore be considered when interpreting the results, since some evaluation metrics, such as model accuracy, may be affected.

Modeling and Evaluation

KNN (K-Nearest Neighbors)

Based on the results obtained for accuracy, which in the context of imbalanced classes was combined with the F1-score because the latter is more precise in this setting, and based on the ROC curves of the individual classes, KNN can be considered a valid option for classifying the data. It appears to provide good classification precision, with an F1-score of 0.87. It is then compared with the other methods to determine which model is actually best suited to this type of data.

Since the ROC curve is designed for binary settings, the three classes in this dataset are adapted to this type of evaluation through a one-vs-all strategy, so that each class is compared with the others. In this way, the three-class classification problem can be treated as a binary problem, where each individual label is compared against the other two combined.

The graphical representation of the ROC curves appears to indicate good data classification. In theory, it is optimal for the curves to be as close as possible to the upper-left corner of the Cartesian plane, and the three curves come very close to this condition. The classification of `Galaxy` is particularly strong among the three curves; this may have been influenced by the fact that this class is the most represented of the three, creating an imbalance in the classification. Considering all the results of this method, the precision obtained is satisfactory and KNN is considered a valid option.

Logistic Regression

For logistic regression, the `glmnet` package was used to implement a multiclass regression model. The model has no penalty; in fact, the value of lambda is zero, so it fits the training data without any restriction on model complexity.

To evaluate model performance and, more specifically, given the lack of penalization, to assess the model's ability not to overfit the data, performance was evaluated on the validation set. Accuracy and F1-score were computed, and through the one-vs-all approach, graphical and numerical indices such as the ROC curve and AUC were implemented to assess performance.

All indices were more than satisfactory, showing overall accuracy above 94% and class-specific sensitivity and specificity indices above 85% for all three classes. In this context, the `STAR` class stands out with a sensitivity of 99.98%, while `QSO` stands out with a specificity of 99.02%.

The ROC curve shows good discriminant ability, as the classifier approaches the upper-left corner for all three classes. This is further supported by the AUC, whose value tends toward 1, again showing the model's strong ability to separate the classes.

Finally, the F1-score is 95%, confirming good classification ability even in the presence of imbalances among the analyzed classes. The unpenalized multiclass logistic regression is therefore considered the best classification model among those proposed, as it does not show evident signs of overfitting. It should be noted that model penalization could be appropriate in settings with greater complexity or a larger number of variables, in order to ensure good generalization.

Decision Trees

LDA - Linear Discriminant Analysis

LDA and QDA are generative models belonging to discriminant analysis. These models differ from discriminative models because they model the distribution of the variables separately for each class. Before applying them, it is necessary to verify whether their assumptions are satisfied.

From the boxplots, it can be observed that for some variables the variance is similar across classes, while for others, such as `log-redshift` or the variables `z` and `g`, clear differences can be seen. This suggests that the assumption of common variance may not be satisfied for all variables.

The graphical analysis of the covariance ellipses suggests that, for some variables, the covariance matrix across classes is similar, supporting the common covariance assumption. However, for other variables, such as `r` and `z`, significant differences can be observed in the size and orientation of the ellipses, indicating that the assumption may not hold for all features. This could affect the effectiveness of LDA, which assumes a common covariance matrix across classes. The results obtained from this model are therefore evaluated and compared with QDA, which does not require this assumption.

Analyzing the QQ-plots of the three classes, the variables do not appear to follow normal distributions. They show significant deviations in the tails relative to the diagonal line used as a reference for normality; some variables, such as `alpha`, also clearly deviate from the theoretical normal distribution in the central part of the distribution. With respect to this third assumption, as with the previous ones, the data do not appear to satisfy the requirements, and therefore the LDA model is not considered appropriate.

The plots show that the variables do not satisfy the condition of class-conditional normality. Graphically, the variable distributions are mostly bimodal and asymmetric, so there is no strong reason to conclude that normality holds.

Since the initial assumptions of normally distributed variables and homoscedastic variance are not verified, LDA is considered inappropriate for classifying the data. The classification evaluation obtained from this model may not be valid, because the assumptions required by the method are not guaranteed in the data to which it is applied. This model also leads to underfitting, since the bias is very high and does not allow for an optimal evaluation. The analysis therefore proceeds with the application of QDA, which does not require the same conditions as the linear model.

QDA - Quadratic Discriminant Analysis

Comparing the confusion matrices for the training set and validation set reveals a clear difference: on the training set, the QDA model appears to perform rather well, while on the validation set the results are clearly disappointing, especially for the `Star` class. This difference is explained by the nature of Quadratic Discriminant Analysis, which tends to overfit the data because it is more flexible than the linear model. For this reason, performance on the training data is almost optimal, whereas when the model is applied to the validation set, the results are poor and the model is not able to generalize sufficiently.

The ROC curve results are not optimal for the classification of `Star`. However, it should be considered that the figure is the result of converting a three-class problem into a binary problem. This change may have influenced the model output, since the QDA model had already shown difficulty in classifying the third category. This issue is probably amplified when the `Galaxy` and `Quasar` classes are combined, since `Star` appears to be classified very poorly against them, almost at the level of random classification around 50%.

Deployment

Given the performance obtained in the previous phase, the analysis proceeds with the logistic regression model, which showed the best ability to discriminate among the classes. The new training set is considered as the union of the previous training and validation sets, which had been used to train the classifier and find the best estimates for the hyperparameters.

The model performance is then evaluated on the test set. Unlike the training and validation sets, which were subject to cleaning and preprocessing, the test set remains closer to its original form and contains new data that the models were not trained to handle. This makes it possible to evaluate the model's generalization ability.

Compared with the results obtained on the validation set, the model shows less satisfactory but still positive results. As explained earlier, the absence of penalization of model complexity reduces the model's ability to generalize, favoring a tendency toward overfitting on the training set and therefore a lower mean squared error than on the test set.

In the confusion matrix, the true positive rate of 44% can be observed, which is the most critical value; however, the other true positive rates are above 68%. The `Star` class has a sensitivity value of 98%. The F1-score is 73%, suggesting that the model can predict the classes correctly with a relatively reliable level of performance.